

KINEMATICS (MOTION IN 1D & 2D)

Class 11 Physics — Complete Comprehensive Concept Notes & Formula Sheet

CBSE Class 11

JEE Main & Advanced

NEET Physics

1. BASIC CONCEPTS & MOTION IN 1D

Kinematics is the branch of mechanics that describes the motion of points, bodies, and systems without considering the forces that cause the motion.

Distance vs. Displacement

Distance	Displacement
Total path length covered by the particle in a given time.	Shortest straight-line distance from initial to final position.
Scalar quantity (always positive or zero).	Vector quantity (can be positive, negative, or zero).
Depends on the path followed.	Independent of path, depends only on initial and final points.
$\text{Distance} \geq \text{Displacement} \Rightarrow \text{Distance} / \text{Displacement} \geq 1$	

Speed and Velocity

- **Average Speed:** Total Distance / Total Time taken = $S_{\text{total}} / \Delta t$
- **Average Velocity:** Total Displacement / Total Time taken = $\Delta \mathbf{r} / \Delta t$
- **Instantaneous Velocity:** Velocity at a specific instant of time:

$$\mathbf{v} = d\mathbf{r} / dt = dx / dt \hat{i} + dy / dt \hat{j} + dz / dt \hat{k}$$

2. EQUATIONS OF MOTION (CONSTANT ACCELERATION)

For motion along a straight line with uniform (constant) acceleration a :

Standard 1D Kinematic Equations

1. $v = u + a t$
2. $s = u t + \frac{1}{2} a t^2$
3. $v^2 = u^2 + 2 a s$
4. $s = ((u + v) / 2) t$
5. **Displacement in the n-th second:** $s_{n\text{-th}} = u + \frac{1}{2} a (2n - 1)$

Motion Under Gravity (Free Fall)

Taking vertically upward as positive ($a = -g$):

- **Maximum Height Reached:** $H_{\text{max}} = u^2 / (2g)$
- **Time of Flight:** $T = 2u / g$ (Time of ascent = Time of descent = u / g)
- **Velocity on re-reaching ground:** $v = -u$ (Magnitude remains u)

3. CALCULUS & GRAPHICAL ANALYSIS

Differential Relationships

$$v = dx / dt$$

Integral Relationships

$$\Delta x = \int v dt \text{ (Area under v-t graph)}$$

$$a = dv / dt = d^2x / dt^2 = v (dv / dx)$$

$$\Delta v = \int a \, dt \text{ (Area under a-t graph)}$$

Key Graphs in 1D Motion

- **Position-Time (x-t) Graph:** Slope = Velocity ($v = dx/dt$).
- **Velocity-Time (v-t) Graph:** Slope = Acceleration ($a = dv/dt$). Area under curve = Displacement (with sign) / Distance (absolute area).
- **Acceleration-Time (a-t) Graph:** Area under curve = Change in velocity (Δv).

4. RELATIVE MOTION IN 1D & 2D

Relative position, velocity, and acceleration of body A with respect to body B:

$$r_{A/B} = r_A - r_B \quad | \quad v_{A/B} = v_A - v_B \quad | \quad a_{A/B} = a_A - a_B$$

Key Relative Motion Problems

- **River-Swimmer / Boat Problem:**
 - *Shortest Path (Crossing directly across river):* Swimmer must head upstream at angle $\sin \theta = v_r / v_{s/r}$. Time $t = d / \sqrt{(v_{s/r})^2 - v_r^2}$.
 - *Shortest Time:* Head perpendicular to riverflow ($\theta = 0^\circ$). Time $t_{min} = d / v_{s/r}$. Drift $x = v_r \times t_{min}$.
- **Rain-Man Problem:** To block rain falling at v_r while running at v_m , hold umbrella along $v_{r/m} = v_r - v_m$. Angle with vertical: $\tan \theta = v_m / v_r$.

5. PROJECTILE MOTION (2D MOTION)

Motion under constant downward acceleration due to gravity g (ignoring air resistance).

Ground-to-Ground Projectile

Initial velocity u at angle θ with horizontal:

- Horizontal Component: $u_x = u \cos \theta$, $a_x = 0$
- Vertical Component: $u_y = u \sin \theta$, $a_y = -g$

Parameter	Formula
Equation of Trajectory	$y = x \tan \theta - (g x^2 / (2 u^2 \cos^2 \theta)) = x \tan \theta (1 - x/R)$
Time of Flight (T)	$T = (2 u \sin \theta) / g$
Maximum Height (H)	$H = (u^2 \sin^2 \theta) / (2g)$
Horizontal Range (R)	$R = (u^2 \sin 2\theta) / g$ (Max range at $\theta = 45^\circ$: $R_{max} = u^2/g$)

Important Projectile Properties

- Complementary angles (θ and $90^\circ - \theta$) yield the same horizontal range R .
- Velocity at maximum height is purely horizontal: $v_{min} = u \cos \theta$.
- Speed at any time t : $v = \sqrt{(u \cos \theta)^2 + (u \sin \theta - gt)^2}$.

6. UNIFORM CIRCULAR MOTION

Motion in a circle of radius R at constant speed v .

- **Angular Velocity:** $\omega = d\theta / dt = 2\pi / T = 2\pi f$
- **Linear-Angular Relation:** $v = \omega R$
- **Centripetal Acceleration:** Directed towards center:

$$a_c = v^2 / R = \omega^2 R = v \omega$$

- **Non-Uniform Circular Motion:** Total acceleration $a = \sqrt{a_c^2 + a_t^2}$, where tangential acceleration $a_t = dv/dt$.

7. QUICK SUMMARY FORMULA SHEET

Displacement in n-th sec:

$$s_n = u + \frac{1}{2} a(2n - 1)$$

Trajectory Equation:

$$y = x \tan \theta (1 - x/R)$$

Time of Flight & Range:

$$T = 2u \sin \theta / g \quad | \quad R = u^2 \sin 2\theta / g$$

Relative Velocity:

$$v_{A/B} = v_A - v_B$$

Centripetal Acceleration:

$$a_c = v^2 / R = \omega^2 R$$